



VELLORE INSTITUTE OF TECHNOLOGY
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
BACSE203 COMPUTER NETWORKS - FALL 2026

Guidelines: Case Study Assignment

1. Submission Requirements:

Submit one combined report per team containing the following three clearly labeled parts:

Part A: Review Paper

Written to the standard of a paper suitable for submission to a reputable conference or journal. It must follow standard academic paper structure and conventions.

Part B: Simulation Based Case Study

Use Cisco Packet Tracer, Wireshark, or Mininet, as specified for your topic. The complete simulation must be documented with relevant screenshots.

Part C: Alignment and Limitations Note

A half page to one page note explicitly connecting the results of Part B with the literature discussed in Part A and explaining what the simulation could not capture.

All three parts are compulsory.

2. Part A: Review Paper Requirements

Format

- Any standard academic paper format is acceptable, such as IEEE, Springer LNCS, ACM, or a clean single column or two column format with clear section headings.
- Consistency is more important than the specific template selected.
- The paper must include the following sections:

Abstract → Keywords → Introduction → Literature Review/Related Work → Comparative Discussion/Analysis → Conclusion and Future Scope → References

- Length: 4 to 6 pages, excluding references.

Literature Requirements

- Include a minimum of 15 references.
- At least 8 references must be peer reviewed research papers or original standards documents, such as RFCs or papers published by IEEE, ACM, or Springer.
- Blogs, Wikipedia, and AI generated summaries should not be used as primary academic references.
- Sources may be obtained from Google Scholar, IEEE Xplore, ACM Digital Library, ResearchGate, or arXiv.
- Every important claim, statistic, or comparison must be supported by an appropriate citation.
- Use one citation style consistently throughout the paper, such as IEEE numbered citations, APA, or another standard academic citation style.

Content Quality

- Organize the literature review thematically rather than source by source.
- Compare and contrast the findings of different studies instead of summarizing each paper one after another.
- Clearly identify areas where the literature:
 - Agrees
 - Disagrees
 - Contains research gaps
- Your own critical analysis must be visible. After discussing the literature, clearly present your own interpretation or inference based on the reviewed studies.

Originality

- The similarity index must be below 15%, excluding references and standard technical terms.
- The report must be written in your own words.
- Heavy copy and paste content or unedited AI generated text will be penalized.

3. Part B: Simulation Case Study Requirements

- Perform the actual simulation or configuration specified for your assigned topic. A symbolic or descriptive explanation without performing the simulation is not acceptable.
- All screenshots must be taken from your own simulation or configuration.
- For every important step, include:
 1. A screenshot of the topology or configuration
 2. A screenshot of the relevant output or result
 3. A short caption explaining what the screenshot demonstrates
- Where comparisons are required, such as before and after results or Condition A versus Condition B, present the results using a small table or a simple chart.
- If the selected tool has a known limitation for the required scenario, state the limitation clearly and honestly.

For example:

- Packet Tracer may not support certain advanced 5G simulations.
- A particular tool may support IPsec but not WireGuard or OpenVPN.

If a limitation requires the use of a simplified approach, explain it clearly. Transparent handling of tool limitations will not be penalized.

4. Part C: Alignment and Limitations Note

This is a new and mandatory section.

In approximately half a page to one page, answer the following questions:

1. How do the results obtained in Part B compare with the findings reported in the literature reviewed in Part A?
 - Do they agree?
 - Do they differ?
 - Are they partially consistent?
 - Provide appropriate reasoning.
2. What real world factors, network scale, environmental conditions, or protocol behaviors could not be captured by the simulation, and why?

3. If a substitute or simplified simulation was used, clearly explain what was simplified and what the simplification leaves out.

For example:

- ACLs used as a simplified representation of Zero Trust access control cannot capture full identity aware authentication and continuous verification.
- Fixed delay values used to represent satellite communication cannot capture all real world variations in satellite links.

5. Evaluation Rubric: 10 Marks

Parameter	What Is Assessed	Marks
Literature Depth and Quality (Part A)	Number and quality of sources, including adequate peer reviewed papers and standards references	2
Critical Synthesis and Organization (Part A)	Thematic organization, comparison across studies rather than sequential summaries, and visible critical judgment	2
Writing Quality and Originality (Part A)	Clarity, structure, consistent citation style, and similarity index below 15%	2
Simulation Execution (Part B)	Correctness and completeness of the actual Packet Tracer, Wireshark, or Mininet task performed	2
Screenshot Documentation (Part B)	Clear, original, and properly captioned screenshots for all relevant steps, with tables or charts where comparisons are required	1
Alignment and Limitations Note (Part C)	Depth and honesty in connecting simulation results with the literature and clearly explaining what the simulation could not capture	1
Individual Viva/Q&A During Seminar	Each member's independent understanding and command over their assigned section	
Total		10.0

6. Submission Checklist

- A standard academic paper structure is used consistently in Part A.
- A minimum of 15 references is included.
- At least 8 references are peer reviewed papers or original standards documents.
- One citation style is used consistently throughout the paper.
- Part A is organized thematically and includes visible critical comparison and analysis.
- Part B includes an actual simulation or configuration performed by the team.
- Original screenshots from the simulation are included with appropriate captions.
- Tables or charts are included wherever comparisons are required.
- Part C includes a substantive Alignment and Limitations Note.
- The similarity index has been checked and is below 15%.
- An Individual Contribution Statement is attached.
- A title page is included with the team name, topic, member names, and registration numbers.